Graduate Certificate in Product Management

The certificate is an online or residential program. The requirements are to complete the core course and two elective courses:

Core Course		
EBGN563	MANAGEMENT OF TECHNOLOGY AND INNOVATION	3.0
Elective Courses		
EBGN553	PROJECT MANAGEMENT	3.0
EBGN565	MARKETING FOR TECHNOLOGY-BASED COMPANIES	3.0
EBGN577	LEADING & MANAGING HIGH PERFORMING TEAMS	3.0

Course substitutions may be approved on a case-by-case basis by the certificate director. Completing the certificate will also position students to complete the master of science in engineering and technology management degree as all the certificate courses can be applied to this degree.

Courses

EBGN502. POLITICAL ECONOMY OF THE ENERGY TRANSITION. 3.0 Semester Hrs.
This course provides an overview of economics, business, and political topics that are commonly found in the energy transition. Many of the assignments relate back to skills that are needed to interact with economics, business, and policy professionals. The course is designed for students with little, if any, social science or business training. Students will build a basic knowledge of economics, finance, and business issues that are relevant to energy markets and industries.
Course Learning Outcomes

- 1. Interpret and assess basic economic intuition and lingo so that one can contribute to projects on the business side
- 2. Evaluate and critique standard investment analysis techniques
- 3. Describe common market structures for natural resource commodities and theorize its impact on firm behavior
- 4. Name the location of basic data on energy price, production, and consumption and demonstrate its evolution over time
- 5. Analyze the politics behind an aspect of the energy transition
- 6. Identify key political actors in the transition
- 7. Design a presentation for the business community that provides a clear value proposition.
- 8. Execute an "elevator pitch" (concise and persuasive speech to spark interest) about an energy/natural resource topic.

EBGN504. ECONOMIC EVALUATION AND INVESTMENT DECISION METHODS. 3.0 Semester Hrs.
Time value of money concepts of present worth, future worth, annual worth, rate of return and break-even analysis are applied to after-tax economic analysis of mineral, petroleum and general investments. Related topics emphasize proper handling of (1) inflation and escalation, (2) leverage (borrowed money), (3) risk adjustment of analysis using expected value concepts, and (4) mutually exclusive alternative analysis and service producing alternatives. Case study analysis of a mineral or petroleum investment situation is required. Students may not take EBGN504 for credit if they have completed EBGN321.

EBGN509. MATHEMATICAL ECONOMICS. 3.0 Semester Hrs.
This course reviews and re-enforces the mathematical and computer tools that are necessary to earn a graduate degree in Mineral Economics. It includes topics from differential and integral calculus; probability and statistics; algebra and matrix algebra; difference equations; and linear, mathematical and dynamic programming. It shows how these tools are applied in an economic and business context with applications taken from the mineral and energy industries. It requires both analytical as well as computer solutions. At the end of the course you will be able to appreciate and apply mathematics for better personal, economic and business decision making. Prerequisites: Principles of Microeconomics, and MATH111.

EBGN510. NATURAL RESOURCE ECONOMICS. 3.0 Semester Hrs.
The threat and theory of resource exhaustion; commodity analysis and the problem of mineral market instability; cartels and the nature of mineral pricing; the environment; government involvement; mineral policy issues; and international mineral trade. This course is designed for entering students in mineral economics. Prerequisite: Principles of Microeconomics.

EBGN511. MICROECONOMICS. 3.0 Semester Hrs.
(I, II, S) This is a graduate course dealing with applied microeconomic theory. The course concentrates on the behavior of individual segments of the economy, the theory of consumer behavior and demand, duality, welfare measures, policy instruments, preferences over time and states of nature, and the fundamentals of game theory. Prerequisites: MATH111, EBGN509. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- No change

EBGN512. MACROECONOMICS. 3.0 Semester Hrs.
This course will provide an introduction to contemporary macroeconomic concepts and analysis. Macroeconomics is the study of the behavior of the economy as an aggregate. Topics include the equilibrium level of inflation, interest rates, unemployment and the growth in national income. The impact of government fiscal and monetary policy on these variables and the business cycle, with particular attention to the effects on the mineral industry. Prerequisites: Principles of Microeconomics, MATH111.

EBGN515. ECONOMICS AND DECISION MAKING. 3.0 Semester Hrs.
The application of microeconomic theory to business strategy. Understanding the horizontal, vertical, and product boundaries of the modern firm. A framework for analyzing the nature and extent of competition in a firm's dynamic business environment. Developing strategies for creating and sustaining competitive advantage.

EBGN521. MICROECONOMICS OF MINERAL AND ENERGY MARKETS. 3.0 Semester Hrs.
(I) This is a graduate course dealing with applied microeconomic theory. This course concentrates on the behavior of the minerals and energy segment of the economy, the theory of production and cost, the theory of consumer behavior and demand, derived demand, price and output level determination by firms, and the competitive structure of product and input markets. Prerequisites: MATH111, EBGN509. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- No change

EBGN523. MINERAL AND ENERGY POLICY. 3.0 Semester Hrs.
(II) An analysis of current topics in the news in mineral and energy issues through the lens of economics. Since many of the topics involve government policy, the course provides instruction related to the